Longest Increasing Subsequence.

1	8	6	4	3	81	42

[1, 8, 81] [1, 8, 42] among others.

DP(i)	1	1	1	1	1	1	1

$\boxed{DP[i] = i \quad \forall\, 0 \le i < n}$

Update as follows.

```
for ( i = 1 ; i < n ; i++) {
    for ( j = 0 ; j < i ; j++) {
        if ( arr[j] < arr[i] && DP(j) + 1 > DP(i))
            DP[i] = DP(j) + 1;
    }
}  return (max( DP));
```

	1	8	6	4	3	81	42
	1	1	1	1	1	1	1
i=1	1	2	1	1	1	1	1
i=2	1	2	2	1	1	1	1
i=3	1	2	2	2	1	1	1
i=4	1	2	2	2	2	1	1
i=5	1	2	2	2	2	3	1
i=6	1	2	2	2	2	3	3

← max(DP)

CamScanner